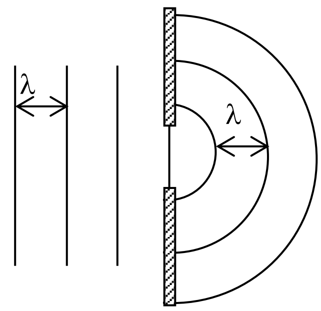
3ai	Diffraction is the <u>spreading of waves</u> when they pass through an opening or round an obstacle. Diffraction effects are the greatest when the <u>width of the opening is comparable with the wavelength of the waves</u>.
3aii	 **Fig. 3.1** Contrast between spreading and curvature Constant wavelength, including drawing of wavefront at the apertures for both figures
3aiii	The wavelength of sound waves is <u>comparable to the size of door gap</u> while the wavelength of light waves is <u>very small compared to the size of door gap</u>. Hence, <u>more significant diffraction</u> is observed for sound than light.
3bi	Intensity at 5.50 m from loudspeaker S_1 = $\frac{P}{4\pi r^2} = \frac{20.0}{4\pi(5.50)^2} = 0.05261$ W m^{-2} Power received by ear = intensity × area of ear = 0.05261 × (2.5 × 10^{-4}) = 1.32 × 10^{-5} W
3bii	Loudspeaker is transmitting energy uniformly in all directions.
3ci	Using $v = f\lambda \Rightarrow \lambda = v / f = 330 / (2.75 \times 10^3)$ = 0.12 m
3cii	Path difference = 4.12 – 3.82 = 0.30 m = 2.5 λ Number of maxima = 2
3ciii	Point A (with path difference = 0) is loud Path difference at point B is 2.5λ hence point B is a soft sound Since intensity ∝ (amplitude of Δpressure)^2, $\frac{I_{A2}}{I_{A1}} = \left(\frac{A_2}{A_1}\right)^2 \Rightarrow \frac{2I}{I} = \left(\frac{A_2}{A_1}\right)^2 \Rightarrow A_2 = \sqrt{2}A_1$ At point B, amplitude of Δpressure, $A_B = \sqrt{2}\,A_1 - A_1$, $\Rightarrow \frac{I_B}{I_{A1}} = \left(\frac{\sqrt{2}A_1 - A_1}{A_1}\right)^2 \Rightarrow I_B = 0.172I$

3civ	The spacing between consecutive maxima increases / Number of maxima decreases
3di	The incident wave is reflected at the wall and a stationary wave is formed. The reflected wave is antiphase with the incident wave at the wall or a displacement node is formed at the wall. (Since displacement node is a pressure antinode,) loud signal at the wall
3dii	$d = 1.5\lambda = 1.5\ (0.12)$ $= 0.18$ m L – Loud region S – Soft region L S L S L S L $\longleftrightarrow$ λ

4a	work done per unit positive charge in bringing